

Durga Pranati Duvvuri

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Education

Georgia Institute of Technology, Atlanta, GA

GPA: 3.78/4.00

B.S. Computer Science (Threads: Modeling/Simulation & Intelligence), Minor: FinTech

Dec 2026

M.S. Computer Science

Jul 2027

Relevant Coursework: Machine Learning, Deep Learning, NLP, Graduate Algorithms, Data & Visual Analytics, Database Systems, Numerical Analysis, Statistics & Applications, Financial Markets & Trading, Finance & Investments, FinTech & Crypto Tokens, Linear Algebra, Differential Equations, Data Structures, Applied Combinatorics

Awards: IEEE EMBS BHI 2025 Data Challenge (2nd Place) | Dean's List (All Semesters) | Faculty Honors

Activities: GT Trading Club | GT Student Finance Society | Women@College Of Computing | GT Board Game Club

Skills & Certifications

Programming: Python, C++, Java, SQL, C, UNIX

ML/Quant: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Time-Series Modeling, Statistical Learning, NLP, Transformers, Hugging Face, Feature Engineering, Text-to-SQL, Backtesting, Regime Analysis, Monte Carlo Simulation, Dimensionality Reduction, Embeddings, Reinforcement Learning

Finance/Trading: Bloomberg Terminal, QuantConnect, Aladdin, Risk Analytics, Financial Sentiment Analysis

Frameworks/Tools: FastAPI, React.js, Django, Git, Excel/VBA

Languages Spoken: English, Telugu, Hindi, Tamil, French

Certifications: Bloomberg Market Concepts, Bloomberg Finance Fundamentals

Work Experience

Technology Intern, BlackRock

Atlanta, GA May 2026 – Aug 2026

- Supporting Aladdin risk analytics platform serving 200+ institutional clients and 100,000+ end users globally across portfolio management, trading, and compliance workflows.
- Applying SQL, Python, and UNIX to ensure accuracy of large analytical datasets and automate data validation processes for investment operations across fixed income, equity, and derivatives.
- Collaborating with cross-functional engineering and client teams to identify data quality issues and improve operational reliability of risk reporting pipelines.

Operational Risk Intern, Freddie Mac — Single Family Division

McLean, VA May 2025 – Jul 2025

- Developed AI-driven operational risk frameworks quantifying exposure across 70+ enterprise processes within the Single Family Division using structured model outputs and scenario analysis.
- Conducted end-to-end risk control mapping across business processes, identifying control gaps and documenting risk mitigation pathways to strengthen the division's operational risk posture.
- Designed and prototyped internal use cases for FredAI, Freddie Mac's in-house LLM, scoping risk-specific applications including automated risk narrative generation and policy Q&A workflows.
- Automated portfolio risk reporting pipelines, reducing manual consolidation overhead and improving executive monitoring frequency; earned 2nd place in company-wide intern case competition.

Teaching Assistant, CS 2340, Georgia Tech

Atlanta, GA Jan 2025 – May 2025

- Mentored 20 students on OOP, software architecture, and Agile principles across semester-long industry-style projects; each group built and evaluated a distinct software system.
- Led coding labs on modular design and testing; reviewed and debugged large codebases with structured feedback on SSDs, class diagrams, sequence diagrams, and iterative delivery cycles.

VIP Researcher, AI for Finance (ML + NLP), Georgia Tech

Atlanta, GA Jan 2025 – May 2026

- Completed 3-semester vertically integrated research program; built Text-to-SQL pipelines over heterogeneous financial schemas (IMF, World Bank, SEC) enabling natural language querying of macro and firm-level time-series data.
- Designed a custom query complexity scoring system combining NL features (schema linkage, lexical density, syntactic depth) and SQL features (operator variety, join depth, aggregation) on a normalized 0–100 scale.
- Benchmarked 7 LLMs (GPT-4o Mini, Gemini 2.0 Flash, DeepSeek V3, LLaMA 3.3, Qwen 2.5 and variants) on execution accuracy, VES, and semantic correctness across ~500 templated financial queries.
- Engineered NLP feature extraction for sentiment, event detection, volatility cues, and liquidity indicators from financial text streams; improved signal classification performance by 12%.

Research Assistant, BIO-MIB Lab, Georgia Tech

Atlanta, GA Aug 2025 – Present

- Built MIST, a unified Text-to-SQL + LLM summarization pipeline over MIMIC-IV EHR data, generating structured patient-level clinical timelines from fragmented multi-table records.
- Evaluated pipeline using ROUGE and BERTScore; ICU Stays and Admissions achieved BERTScore F1 > 0.81; identified degradation on temporally complex tables, motivating schema-aware grounding improvements.
- Presented at Georgia Tech STAR AI and UROP Symposiums; findings on query-guided summarization reducing hallucination in clinical LLM outputs are being prepared for publication.

Business Analyst Intern, Living Full Out

Santa Monica, CA Sep 2024 – Feb 2025

- Built data pipelines across 150+ Excel datasets tracking 2,000+ clients; improved engagement conversion rates by 17% through predictive analytics.
- Supported operational analytics and coordinated logistics for 300+ guest workflows on nationally syndicated radio show.

Full-Stack Developer Intern, iQuadra Information Services

Atlanta, GA Aug 2023 – Aug 2024

- Built BugSprint, a React + Django full-stack bug tracking system for real-time workflow management.
- Improved backend performance and system throughput by 30% through optimization of API and database operations.

Projects

Market Pulse: SOFR Forecasting & Analysis

Aug 2025 – Present

- Developed time-series forecasting pipelines for SOFR rate movements using statistical and ML models; evaluated forecast stability via temporal cross-validation and forecasting error decomposition.
- Built regime-aware rolling-window evaluation framework to quantify forecast degradation and macro-driven structural breaks in SOFR dynamics.

Stock Similarity via Representation Learning

Jan 2025 – May 2026

- Benchmarked 8 embedding methods (PCA, ICA, Kernel PCA, Isomap, LLE, MAE, VAE, PatchTST) on stock return time-series; evaluated PatchTST vs. manifold methods to rank cross-security similarity.